

Introduction to Optimization: Part 1

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What is optimization? An example

“A factory makes tables and chairs. Each table earns \$20 profit and requires 1 hour of labor and 10 units of wood. Each chair earns \$30 profit and requires 2 hours of labor and 5 units of wood. The factory has 30 hours of labor and 120 units of wood available per week. How many tables and chairs should they produce to maximize profit?”

- What decisions do we need to make?
- What limitations do we face?
- What do we want to optimize?

Optimization helps make decisions under various constraints!

Three ingredients of an optimization problem

- **Decision variables:** encodes the choices to be made
 (x_1, x_2) : number of tables / chairs to produce
- **Constraints:** these are equations/inequalities that *link* different decisions together
 - Labor: $x_1 + 2x_2 \leq 30$
 - Wood: $10x_1 + 5x_2 \leq 120$
 - Non-negativity: $x_1, x_2 \geq 0$
- **Objective:** this is an expression to either minimize or maximize
 $\max 20x_1 + 30x_2$

Three ingredients of an optimization problem (cont.)

Mathematically we can write this as:

$$\begin{array}{ll}\max & 20x_1 + 30x_2 \\ \text{such that} & x_1 + 2x_2 \leq 30 \\ & 10x_1 + 5x_2 \leq 120 \\ & x_1, x_2 \geq 0\end{array}\qquad \begin{array}{ll}\max & \mathbf{c}^\top \mathbf{x} \\ \text{such that} & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}\end{array}$$

Here, $\mathbf{c} = \begin{pmatrix} 20 \\ 30 \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 10 & 5 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 30 \\ 120 \end{pmatrix}$, and $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.

Since the objective is linear, and the constraints are linear, we have a *linear* optimization (LO) problem!

(Also called a *linear program* (LP))

(Aside: different forms of an optimization problem)

- Sometimes you will see a minimization objective instead of a maximization objective
- Sometimes, you can get a mix of $\leq$, $=$, and $\geq$
- **Standard form** of an optimization problem is when you write it as:

$$\begin{aligned} \min \quad & \mathbf{c}^\top \mathbf{x} \\ \text{such that} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

(Aside: other types of optimization problems)

We had real-valued decision variables, a linear objective and linear constraints in linear optimization. Other types of optimization include:

- Integer (or mixed-integer) optimization: when all (or some) of the decision variables are integer (i.e. $x \in \mathbb{Z}$ instead of $x \in \mathbb{R}$)
- Nonlinear optimization: when either the constraints or objective are nonlinear in the decision variables:

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ \text{such that} \quad & g_i(\mathbf{x}) \leq 0, \quad \forall i \in [m] \\ & \mathbf{x} \geq 0 \end{aligned}$$

- Convex optimization: special case where f and g_i are convex in x
- Conic optimization: when $\mathbf{x} \geq 0$ is replaced by $\mathbf{x} \in \mathcal{K}$ for a cone $\mathcal{K}$ (out of scope)

(Aside: convexity)

Convex Set: A set $C \subseteq \mathbb{R}^n$ is *convex* if for any $x, y \in C$ and any $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in C.$$



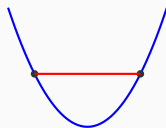
Convex



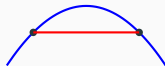
Non-convex

Convex Function: A function $f : C \rightarrow \mathbb{R}$ is *convex* if for any $x, y \in C$ and any $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$



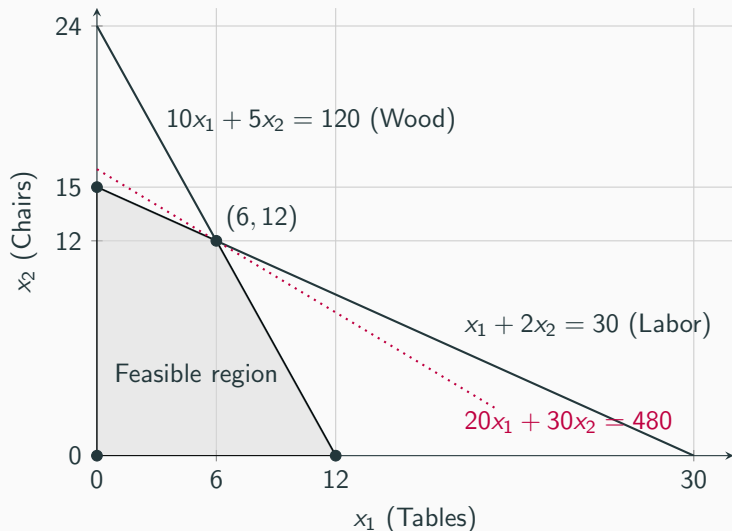
Convex



Non-convex

Visualizing the feasible region

Feasible region: the points $\mathbf{x} \in \mathbb{R}^n$ that satisfy the constraints



Corners of the feasible region

- The feasible region of the LO problem is a **polyhedron**

A polyhedron is a finite intersection of half-spaces:

$$\begin{aligned}\text{Polyhedron} &= \{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{Ax} \leq \mathbf{b} \in \mathbb{R}^m \} \\ &= \left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_i^\top \mathbf{x} \leq b_i, \forall i \in [m] \right\} \\ &= \underbrace{\left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_1^\top \mathbf{x} \leq b_1 \right\}}_{\text{Halfspace 1}} \cap \cdots \cap \underbrace{\left\{ \mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}_m^\top \mathbf{x} \leq b_m \right\}}_{\text{Halfspace } m}\end{aligned}$$

- This shaded region (most of the time) has **corners**, and the corners end up being optimal for the optimization problem!
- That means that solving linear optimization can be done via searching over the corners, instead of over the whole feasible region

Julia demo